URBANFLOW: Fifteen-Minute NYC Yellow Taxi Pickup-Demand Forecasting with Gradient-Boosted Trees

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***Abstract*—**This paper presents URBANFLOW, a zone-level model of New York City yellow taxi pickup demand at 15-minute resolution. Public 2025 taxi trip records were converted into a complete panel spanning 262 taxi zones and 35,040 nominal 15-minute intervals. The 48,613,289 accepted pickup events generated 9,180,480 zone-time rows; after a seven-day historical warm-up, 9,004,416 rows remained for supervised forecasting. Four LightGBM variants were examined using chronological training and validation splits. On the held-out November–December 2025 test period (1,534,272 observations), the selected L2-objective model achieved mean absolute error (MAE) of 1.330524, compared with 1.580073 for previous-interval persistence, a relative reduction of 15.79%. A paired seven-day block bootstrap gave a 95% interval of 14.04%–17.32% for this reduction. Performance varied strongly by zone activity: 178 of 262 zones improved, including all 79 high-activity zones, whereas 69 zones below 1,000 pickups did not improve. A publicly deployed dashboard replays precomputed test predictions. These findings characterize historical one-step-ahead forecasting in 2025, not live inference or out-of-year generalization.

***Index Terms*—**Taxi demand forecasting, time-series regression, LightGBM, urban mobility, DuckDB, model evaluation, spatial heterogeneity.

I. Introduction

Short-horizon pickup-demand estimates are relevant to urban mobility analysis, fleet allocation and demand monitoring. Operational demand is spatially heterogeneous: dense pickup zones exhibit sustained demand while many others contain long sequences of zero-count intervals. This creates a challenging regression setting in which a low overall error can coexist with poor reliability for individual regions.

URBANFLOW investigates whether gradient-boosted decision trees, using observed historical demand and time-of-day descriptors, reduce forecasting error compared with simple carry-forward baselines. The contribution is an applied, carefully evaluated 2025 case study with explicit data accounting, chronological model selection, holdout evaluation and zone-stratified diagnostics. It does not claim a new forecasting architecture or superiority over specialist methods not tested here.

The primary research question is: *Can a LightGBM model reduce one-step-ahead 15-minute pickup-count error across 262 NYC taxi zones relative to historical-demand baselines, and where do the gains fail?*

TABLE I
Data accounting for the 2025 study

Quantity	Count
Accepted 2025 pickup events	48,613,289
Full-year zone-time rows	9,180,480
Feature-engineered rows	9,004,416
Pickups in retained prediction targets	47,990,987
Pickups in the seven-day warm-up	622,302
Zero-count rows in final panel	5,262,694

II. Data Provenance and Panel Construction

A. Trip records and spatial scope

The input consists of monthly 2025 New York City Taxi and Limousine Commission (TLC) yellow taxi trip records and the accompanying taxi-zone lookup [1]. Pickup timestamps and pickup-zone identifiers form the essential raw fields. Zones were restricted to those associated with the five NYC boroughs, yielding 262 retained zones in the executed pipeline. The trip files may include reporting inaccuracies, as described by their publisher [1].

B. Temporal aggregation

For zone z and 15-minute bin t, the target is

$$y_{z,t} = \sum_{i=1}^{N} \mathbb{1}\{z_i = z,\ \tau_i \in B_t\}, \tag{1}$$

where z_i and τ_i denote the pickup zone and pickup timestamp of accepted trip i, and B_t is the time interval indexed by t. DuckDB groups records by pickup zone and time bucket, then left-joins counts onto a complete grid of zone-time pairs. Missing combinations are assigned count zero, rather than discarded.

The nominal full-year grid has

$$262 \times 96 \times 365 = 9{,}180{,}480 \text{ rows}, \tag{2}$$

where each day has 96 fixed 15-minute bins. Seven days of initial historical context are required for the weekly lag. The retained supervised panel therefore contains

$$262 \times 96 \times 358 = 9{,}004{,}416 \text{ rows}. \tag{3}$$

TABLE II
MODEL COMPARISONS ON 2025 VALIDATION DATA

Method	MAE	High-demand MAE	Peak F1
V1: L1	1.3159	9.6884	0.5553
V2: L2	**1.3117**	**8.7333**	**0.6592**
V3: Poisson	1.3178	9.0417	0.6215
V4: weighted L1	1.3165	9.2674	0.5690
Persistence	1.5882	10.8565	0.6241

Note: High demand is an observed count of at least 37. The shown peak F1 comparison uses a common threshold of 150; the subsequently selected alert operating point is reported separately.

Table I distinguishes individual pickup events from aggregated forecasting rows. Many trip records contribute to a single observed count; the difference between 48.6 million events and 9.0 million model rows does not indicate loss of the underlying pickup-volume information. The first seven days contribute historical features but are excluded as prediction targets.

III. Forecasting Features and Experimental Design

A. Predictor construction

The 11 model predictors are the categorical `zone_id`; `hour`, `minute`, `day_of_week`, and `is_weekend`; demand lags at 1, 4, 96 and 672 prior bins; and rolling means over the previous 4 and 96 bins. For a history length w, the rolling mean is

$$\bar{y}_{z,t}^{(w)} = \frac{1}{w}\sum_{j=1}^{w} y_{z,t-j}, \quad w \in \{4, 96\}. \tag{4}$$

The SQL window specifications end at `1 PRECEDING`; thus neither rolling mean includes the current target. Similarly, lag features are taken from previous rows of the same zone. The feature Parquet also contains timestamp and month fields, but month is not among the 11 inputs of the frozen model.

B. Chronological partitioning

The complete feature panel was split by timestamp, not randomly. The training sample spans 8 January–31 August 2025 (5,935,872 rows); validation spans 1 September–31 October (1,534,272 rows); and held-out testing spans 1 November–31 December (1,534,272 rows). The seven preceding days provide lag context. Model objectives and alert thresholds were selected using validation results, after which final test evaluation used the frozen selections.

C. Regressor and selection criterion

The original project compared four LightGBM configurations: V1 with absolute-error regression, V2 with squared-error regression, V3 with Poisson loss, and V4 with weighted absolute-error regression. LightGBM is a gradient-boosting framework for tree-based learning [2]. Table II shows that V2 had the lowest validation MAE and high-demand MAE among these variants, so V2 was retained for the frozen test. Baseline metrics were recomputed from aligned observations rather than relying on an earlier hard-coded preliminary value.

TABLE III
HELD-OUT TEST PERFORMANCE, 1 NOV.–31 DEC. 2025

Method	MAE	RMSE	WAPE (%)
LightGBM V2	**1.330524**	**3.331967**	**24.1029**
15-min persistence	1.580073	4.053617	28.6236
Previous day	2.216162	6.751606	40.1465
Previous week	2.232044	6.743614	40.4342

D. Forecasting information boundary

For each test interval the model uses *realized* preceding pickup counts, including prior test-period intervals. Consequently, the evaluation describes a rolling, one-step-ahead historical replay. It is not a fixed forecast of the entire November–December period issued at the end of October, nor an operational service ingesting live measurements.

IV. Held-Out Test Results

A. Metrics and reference methods

For n observations, with actual count y_i and forecast $\hat{y}_i$, the evaluation computes

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|, \tag{5}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}, \tag{6}$$

$$\mathrm{WAPE} = 100\frac{\sum_i |y_i - \hat{y}_i|}{\sum_i y_i}. \tag{7}$$

Three baselines were tested: the preceding 15-minute observation, the same interval on the preceding nominal day, and the same interval on the preceding nominal week. The saved V2 predictions were clipped as $\hat{y}_i \leftarrow \max(0, \hat{y}_i)$ to enforce nonnegative demand. The verification outputs identified 36,572 negative raw model predictions prior to clipping.

The relative MAE reduction versus persistence is

$$\Delta_{\mathrm{MAE}} = 100\left(1 - \frac{1.330524}{1.580073}\right) = 15.7935\%. \tag{8}$$

The evaluation result in Table III is an advantage over the stated baselines, not evidence of dominance over other modern models that were not evaluated.

V. Temporal and Geographic Reliability

Across the 61 test days, the model outperformed persistence on 60 days when errors were aggregated by day. At the zone level, MAE improved for 178 of the 262 retained zones. These averages conceal substantial variation by realized demand volume.

The 69 zones with fewer than 1,000 pickups during the test period all had higher model MAE than persistence, whereas all 79 high-activity zones improved (Table IV). A substantial portion of the feature panel has no observed pickups (5,262,694 of 9,004,416 rows, approximately 58.44%). In such sparse series, even small nonzero predictions can exceed the error of a

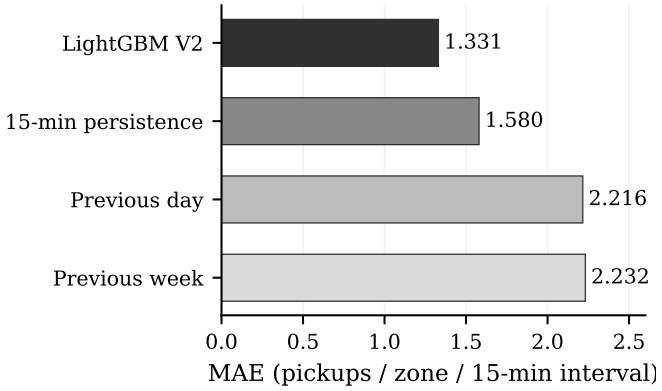


Fig. 1. Held-out test MAE of LightGBM V2 and historical-demand baselines. Lower is better.

TABLE IV
ZONE-LEVEL RELIABILITY BY TEST-PERIOD ACTIVITY

Pickups in test	Improved	Model MAE	Base MAE
0–99	0 / 30	0.0345	0.0097
100–999	0 / 39	0.1631	0.1480
1,000–9,999	99 / 114	0.5449	0.5987
10,000+	79 / 79	3.5327	4.2995

persistence forecast of zero. The test-period activity groups are descriptive, *not* valid deployment segmentation rules, because their construction uses future test counts.

VI. Uncertainty and Peak-Demand Diagnostics

A. Paired temporal block bootstrap

To account for within-week temporal dependence, the executed evaluation aggregated paired model and baseline absolute errors by test day and resampled the 61-day period using circular blocks of seven days. With 5,000 bootstrap replications and random seed 42, the percentile 95% interval for the relative MAE reduction in Eq. (8) was 14.0428%–17.3151%. The interval describes variability under this resampling scheme within the available test period; it does not guarantee performance in a different calendar year.

B. Peak-demand classification

A separate retrospective peak analysis defined a positive event as at least 150 observed pickups in one zone and 15-minute interval. Prediction cutoffs were fixed from validation (145 for the model and persistence in the saved configuration) before testing. The model yielded precision 0.70685, recall 0.63967 and F1 0.67158; persistence yielded precision 0.57267, recall 0.70992 and F1 0.63395. Thus V2 improved test precision and F1 at the selected operating point, but had lower recall. This trade-off is important if missed demand spikes are operationally expensive.

VII. Reproducibility and Public Demonstration

The research materials comprise the original Colab notebook, reorganized data-processing and model-development notebooks, executed final evaluation, and retained historical investigations.

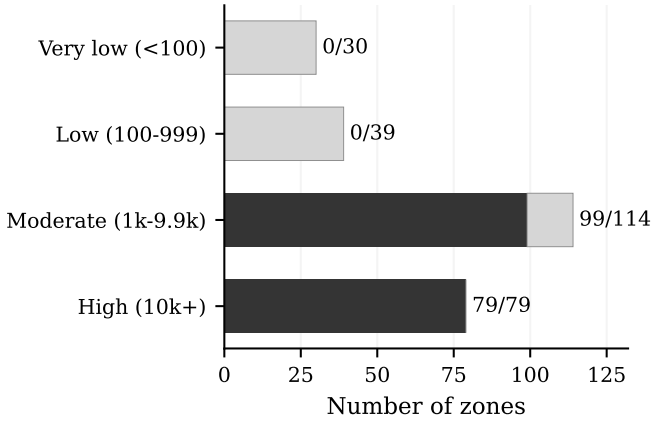


Fig. 2. Zone-level comparison by realized held-out pickup volume. Dark bars indicate zones improved over persistence; light bars indicate zones not improved.

The saved model bundle contains the LightGBM estimator, ordered feature names and taxi-zone categorical encoding. Reproducing the reported test MAE requires the same model bundle, saved feature Parquet, chronological filtering, categorical ordering and nonnegative prediction clipping. New experimental changes should be assessed on new validation data, not iteratively tuned against the frozen test period.

A Streamlit application displays precomputed November–December 2025 predictions, historical demand curves, baseline comparisons, zone/day selectors and CSV exports. Its approximately 9.83 MiB dashboard data file is an artifact of test predictions, not a live trip feed. Project materials are available at https://github.com/div12-space/URBANFLOW-NYC-Taxi-Forecasting.

The numbers presented here are drawn from the author's supplied, executed notebooks and their saved outputs. Reloaded-model test metrics were reproduced in an executed verification notebook. Preparation of this manuscript did *not* independently rerun the full monthly raw-data ingestion and four-model training pipeline.

VIII. Limitations and Future Work

Single-year coverage. All model development and evaluation are within 2025, limiting claims about 2026 or later. An out-of-year test is a higher-value extension than increasing the number of trip-level rows for the same 15-minute forecasting objective.

Clock-time treatment. The source uses timezone-naive timestamps and a fixed 96-bin daily grid. Ambiguities around New York daylight-saving transitions are not fully resolved by the present processing pipeline.

Sparse-demand failure. Zone-level analysis reveals systematic weaknesses in low-volume zones. A zero-aware or specialized sparse predictor is a reasonable follow-up, provided that both its design and decision thresholds are chosen without using the final test set.

Operational validity. The current results assume that the preceding observed count and lag features are available at

prediction time. Real-time ingestion, delays, missing features and recursive multi-step forecasting have not been evaluated.

Benchmark scope and record quality. Comparisons are limited to the evaluated LightGBM configurations and three simple demand baselines. The results do not establish superiority over dedicated spatiotemporal or probabilistic forecasting models, and the TLC source records have known collection-quality caveats [1].

Future work should prioritize independent environment-level reproducibility, out-of-year validation, explicit time-zone handling, sparse-zone calibration and the operational assessment of prediction latency.

IX. Conclusion

URBANFLOW demonstrates a historical zone-level pipeline for short-horizon yellow taxi demand estimation using 2025 NYC TLC records. Its selected LightGBM V2 model achieved 1.330524 test MAE and a 15.79% reduction relative to previous-interval persistence on a 61-day held-out period. The method is consistently advantageous in high-activity zones, but weaker in low-volume zones. The value of the study lies in its data accounting, chronological evaluation, transparent limitations and reproducible frozen-model evidence, rather than a claim of a novel forecasting algorithm.

References

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